Impeller peripheral speed definition

Index

1. Purpose & Scope 3

2. Definitions 3

3. General Procedure - Description 3

4. Description of the methodology 4

4.1. Dimensioning of the maximum peripheral speed limit definition 4

4.2. High energy criteria 4

4.3. Material and fluid selection criteria 5

4.3.1. Demineralized water 6

4.3.2. Slurry fluids 6

4.3.3. Seawater and produced water 7

4.3.4. Hydrocarbons 7

5. Description of the module 12

6. Acceptance criteria 13

7. Limitations 14

8. References 14

1. Purpose & Scope

This procedure establishes the criteria to calculate the maximum peripheral speed of impeller by means of Gulich [1] formulations.

This procedure is applicable to open and closed impellers designed in Nova Milanese and La Spezia plants.

2. Definitions

n:	Rotational speed of the pump rotor	[rpm]
Hbep@D2max:	Pump head at best efficiency point at D2max	[m or ft]
Qbep:	Flowrate at best efficiency point	[m^3/h or GPM]
D2max:	Maximum external diameter	[mm or in]
Nq:	Pump specific speed	
U2:	Impeller peripheral velocity	[m/s]

3. General Procedure - Description

This procedure is intended to support the pump designer on the definition of the maximum allowable peripheral velocity of impellers by using the **Mod.559** tool.

The main parameters that affect speed limit are:

- Material
- Pumped fluid
- Operative conditions

Mod.559 spreadsheet is an empirical procedure that should be used in two different ways:

- Definition of the maximum allowable peripheral speed of an impeller.
- Check of peripheral speed for an established impeller D2max.

4. Description of the methodology

The following sections describe the methodology used in the **Mod.559**.

4.1. Dimensioning of the maximum peripheral speed limit definition

The maximum peripheral speed definition is based on the formulation proposed by [1] on tables 14.6 14.17 and the high energy definition contained on Chapter 15. The method evaluates the maximum allowable peripheral velocity for all centrifugal pumps, which is the tangential velocity at the outermost edge of an impeller as it rotates. This critical parameter is determined by the impeller diameter and rotational speed, calculated using the formula:

$$U_2 = \frac{n}{60} * 2\pi * \frac{D2_{max}}{2000}$$

The maximum peripheral speed an impeller can safely operate at is fundamentally limited by the tensile strength of the impeller's material and by the properties of the handled fluid and dispersed particles.

The primary mechanical stress on impellers at high speeds originates from centrifugal forces rather than aerodynamic stresses.

The erosion rate in pumps handling suspended solids follows a cubic relationship with particle velocity:

$$\text{Erosion} \propto C^3$$

Since particle velocity is directly related to impellers tip speed, its reduction has a dramatic impact on erosive wear. For slurry applications, material class and impeller diameter significantly influence maximum allowable speeds.

4.2. High energy criteria

According to Gulich [1], pumps are categorized into three groups based on their energy level, with G1 representing the highest energy achievable in industrial applications. *Figure 1* displays curves of peripheral speed versus specific speed (Nq); the top curve indicates the maximum peripheral speed for high energy pumps.

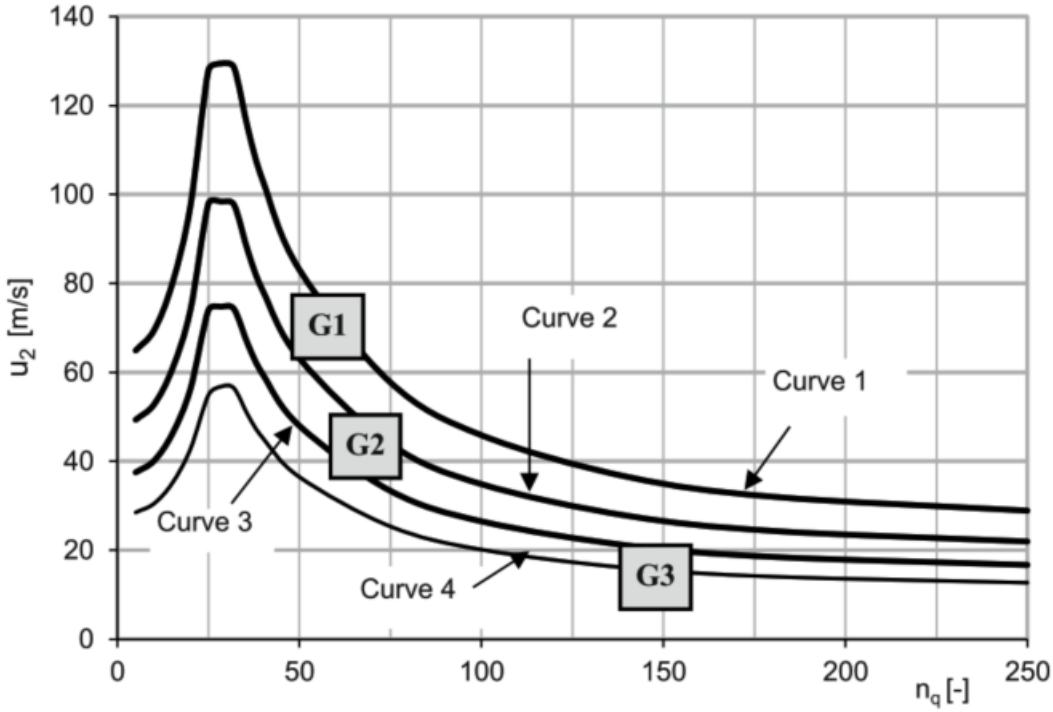


Figure 1: energy level classifications based on the maximum peripheral speed of impellers.

4.3. Material and fluid selection criteria

The combination between impeller's material and pumped fluid characteristics greatly affects the maximum achievable peripheral speed.

Typical fluids handled by pumps manufactured by the Organization are:

- Potable water
- Demineralized water
- Hydrocarbon
- Seawater
- Produced water
- Slurry

Typical cast materials selected for the impellers used by the Organization are:

- Cast Iron A278-30
- Nodular Cast Iron A536-60-40-18
- Ni Resist A439 Type D2

- Cast Steel A216 Gr WCB
- Martensitic A487/A743 CA6NM Gr A
- Martensitic A487/A743 CA15 Gr D
- Austenitic A451/A743 CF-3
- Austenitic A451/A743 CF-3M
- Duplex A995/A890 1B
- Duplex A995/A890 4A CD3MN
- Duplex A995 3A
- SuperDuplex A995/A890 6A CD3MWCuN
- SuperDuplex A995/A890 5A CE3MN
- B584 C86300
- B427 C90700
- C148 958
- Hastelloy CA574 N10276

The effects of fluid characteristics on pump materials are discussed below:

4.3.1. Demineralized water

Demineralized water is considered most aggressive than potable water due to the lower ionic content. This makes it chemically aggressive because it tends to leach ions from metal surfaces to reach equilibrium, accelerating corrosion of impeller materials, especially at high velocities where protective oxide layers can be stripped away. This can dissolve protective films and coatings faster. Combined with high-speed flow, this accelerates erosion-corrosion synergy.

4.3.2. Slurry fluids

Slurry fluids contain suspended solid particles that create hydro abrasive wear on impeller surfaces. For those fluids, the formulation below proposed by Gulich [1] is used, considering dimensions and mass concentrations of suspended solids:

$$u_{2.al} = u_{2.ref}\left\{1 - 0.44\left(\frac{x\, d_s}{(1-x)\, d_{ref}}\right)^{0.33}\right\}$$

Where:

- u2.ref = 47 [m/s] for metallic materials, 31 [m/s] for rubber material
- dref = Reference diameter 1 [mm]
- x = mass concentrations of solid particles
- ds = solid particles average diameter [mm]

4.3.3. Seawater and produced water

Sea water typically contains dissolved salts, like sodium and magnesium chlorides. In coastal and port areas, sea water can be contaminated with particularly aggressive substances such as hydrogen sulphide (H_2S).

When selecting materials for sea water applications, it is important to consider these potential contaminants.

Produced water from oil wells can have much higher salt content and is usually weakly acidic, often containing dissolved H_2S but lacking oxygen.

The behavior of stainless steels in sea water varies with flow velocity: at low or no flow, there is a high risk of local corrosion due to limited oxygen; above 1–2 m/s, increased oxygen transport passivates the metal surface; but above a certain threshold velocity, erosion corrosion can destroy the protective film, especially in sensitive materials like austenitic steels.

Stainless steels are generally resistant to uniform corrosion but may still suffer from localized forms like pitting, crevice corrosion, and deposit-related attacks. Austenitic steels without molybdenum (AISI 304) are suitable for cold water with chloride content up to 200 ppm, while those with molybdenum (AISI 316) can handle up to 1,000 ppm.

4.3.4. Hydrocarbons

Hydrocarbons have lower density and better lubricating properties than water, which lead to reduced mechanical and hydraulic stresses on pump components. This, in turn, permits higher rotational velocities before reaching material stress or cavitation limits. However, the module considers the same speed limit for hydrocarbon and potable water.

According to API 610 standard [2], the maximum recommended impeller tip speed on overhung pumps is 62 m/s, unless specific approval is granted by the purchaser.

Based on tables 14.6 and 14.17 of Gulich [1], a velocity limit is provided based on the impeller's material and the handled fluid.

Velocity limit is calculated using the formula below:

$$u_{2.al} = \frac{w_{2.al}}{w_2^*}$$

Where:

- U2.al is the allowable impeller peripheral speed.
- W2.al is a function of the material and fluid as shown in *Figure 3*, *Figure 4* and *Figure 5.*
- W2* is the peripheral velocity ratio w2u* proposed by Gulich[1] as shown in *Figure 2.*

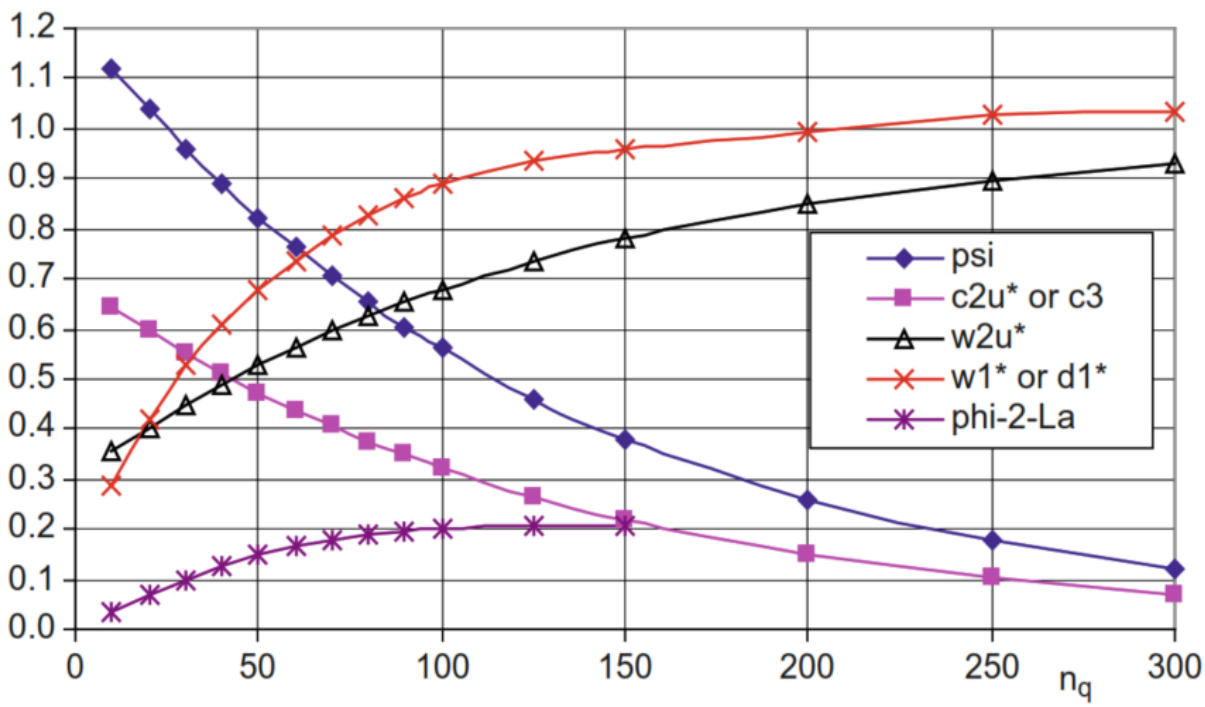


Figure 2. statistical pump characteristics as per Gulich [1].

Table 14.6 (1) Materials for sea water at T < 30 °C				$H_{al} \approx \frac{1}{5} w_{al}^2$ at $\psi_{opt} \approx 1$
Type of material	**Sea water**	**Natural**		**Polluted**
	pH	> 5.5		4 to 5.5
	O_2-concentration	> 1 ppm	< 0.01 ppm	>1 ppm
	H_2S-concentration	≈ 0 ppm	< 10 ppm	Up to 500 ppm
Carbon steels, GS-C25, cast iron (GG-25, GGG-40)	UC (w < 0.5 m/s)	$c_{cor} \approx 0.1$ mm/year		Not applicable, heavy attack in a short time
	Erosion corrosion	$c_{cor} \approx 0.15\ w_r^{1.2}$ mm/year		
	Application limit	$w_{al} \approx 2$ m/s		
	LC at w < 1 m/s	$c_{LC} \approx 0.3$ to 0.8 mm/year		
	SC (GG, GGG)	Graphitization		
Ni-Resist Type GGG-NiCr 20 2	UC (w < 0.5 m/s)	$c_{cor} \approx 0.07$ mm/year		
	Erosion corrosion	$c_{cor} \approx 0.031 w_r^{0.95}$ mm/year		
	Application limit	$w_{al} \approx 20$ m/s	H = 80 m	
	LC at w < 1 m/s	$c_{LC} \approx 0.1$ mm/year		
	SC	Resistant		
Martensitic Cr-Ni-steels, Type 1.4317	UC (w < 0.5 m/s)	$c_{cor} \approx 0.1$ mm/year		
	Erosion corrosion	$c_{cor} \approx 10^{-6}\ w_r^{3.8}$ mm/year (?)		
	Application limit	*Not recommended*		
	LC at w < 1 m/s	$c_{LC} \approx 1.5$ mm/year		
Austenitic Cr-Ni-steels, Type 1.4409	UC (w < 0.5 m/s)	$c_{cor} \approx 0.01$ mm/year		
	Erosion corrosion	Fairly resistant		Limits application
	Application limit	$w_{al} \approx 40$ m/s	H = 300 m	$w_{al} \approx 25$ m/s; H = 120 m
	LC at w < 1 m/s *very sensitive to LC*	10%Ni: $c_{LC} \approx 1.8$ mm/year 29%Ni: $c_{LC} \approx 0.2$ mm/year *Not recommended.*		Application not recommended
	SC	To be avoided with appropriate alloy: Nb, Ti, C < 0.03		
Super-austenitic Ni-Cr-steels, Ni > 25% Type 1.4587	UC (w < 0.5 m/s)	$c_{cor} \approx 0.01$ mm/year		
	Erosion corrosion	Fairly resistant		
	Application limit	$w_{al} \approx 60$ m/s		$w_{al} \approx 55$ m/s
	LC at w < 1 m/s			
	SC	To be avoided with appropriate alloy: Nb, Ti, C < 0.03		
Duplex-steels	UC (w < 0.5 m/s)	$c_{cor} \approx 0.01$ mm/year		Suitable with PI > 40, for definition of PI see Eq. (14.8)
	Erosion corrosion	Fairly resistant		
	Application limit	$w_{al} \approx 55$ m/s	H = 600 m	
	LC at w < 1 m/s	$c_{LC} \approx 0$ mm/year		
UC: Uniform corrosion LC: Local corrosion (crevices, pitting) SC: Selective corrosion c_{LC}: Pitting velocity		$w_r \equiv w/w_{Ref}$ with $w_{Ref} = 1$ m/s		
w_{al} is the velocity relative to the component ($w \neq u$); w_1*, c_{2u}* can be estimated from Fig. 3.25				

Figure 3: velocity ratio definition for several impeller materials (1).

Table 14.6 (2) Materials for sea water at T < 30 °C		$H_{al} \approx \frac{1}{5} w_{al}^2$ at $\psi_{opt} \approx 1$		
Type of material	**Sea water**	**Natural**		**Polluted**
	pH	> 5.5		4 to 5.5
	O_2-Concentration	> 1 ppm	< 0.01 ppm	>1 ppm
	H_2S-Concentration	≈ 0 ppm	< 10 ppm	Up to 500 ppm
Tin bronze Copper-Tin-Alloys G-CuSn 12 2.1052.01	UC (w < 0.5 m/s)	$c_{cor} \approx 0.03$ mm/year		Not applicable, heavy attack in a short time
	Erosion corrosion	$c_{cor} \approx 10^{-5} w_r^3$ mm/year		
	Metal loss increases with ammonia content			
	Application limit	$w_{al} \approx 23$ m/s	H = 100 m	
	LC at w < 1 m/s	$c_{LC} \approx 0.25$ mm/year		
Copper-tin-zinc-alloy G-CuSn 5ZnPb 2.1096.01	UC (w < 0.5 m/s)	$c_{cor} \approx 0.03$ mm/year		
	Erosion corrosion	$c_{cor} \approx 0.06\, w_r^{0.95}$ mm/year		
	Metal loss increases with ammonia content			
	Application limit	$w_{al} \approx 10$ m/s	H = 25 m	
Aluminum-bronze G-CuAl 10Ni 2.0975.01	UC (w < 0.5 m/s)	$c_{cor} \approx 0.05$ mm/year		
	Erosion corrosion	$c_{cor} \approx 3.6\times10^{-4} w_r^{1.8}$ mm/year		
	Metal loss increases with ammonia content			
	Application limit	$w_{al} \approx 30$ m/s	H = 180 m	
	LC at w < 1 m/s	$c_{LC} \approx 0.2$ mm/year		
	SC	With ≥ 4% Ni no SC		
Nickel base alloy NiCu30Al 2.4374	UC (w < 0.5 m/s)	$c_{cor} \approx 0.03$ mm/year		–
	Erosion corrosion	Fairly resistant		–
	Application limit	$w_{al} \approx 50$ m/s	H = 500 m	$w_{al} \approx 5$ m/s
	LC at w < 1 m/s	$c_{LC} \approx 0.4$ mm/year		–
Hastelloy C NiMo16Cr15W 2.4819	UC (w < 0.5 m/s)	$c_{cor} \approx 0.01$ mm/year		–
	Erosion corrosion	Resistant		–
	Application limit	$w_{al} \approx 60$ m/s	H = 700 m	$w_{al} \approx 60$ m/s H = 700 m
	LC at w < 1 m/s	$c_{LC} \approx 0$ mm/year		–
UC: Uniform corrosion LC: Local corrosion (crevices, pitting) SC: selective corrosion c_{LC}: Pitting velocity		$w_r \equiv w/w_{Ref}$ with $w_{Ref} = 1$ m/s		
w_{al} is the velocity relative to the component ($w \neq u$); w_1*, c_{2e}* can be estimated from Fig. 3.25				

Figure 4: velocity ratio definition for several impeller materials (2).

Table 14.17 Velocity limits for cast materials $H_{al} \approx 0.2 \times w_{al}^2$ at $\psi_{opt} \approx 1$

		DIN 17006	GG-25, EN-GIL-250	GGG-40, EN-GJS-400-15	GGG-NiCr 20 2	GS-C25, GP240GH	GX8CrNi12	GX4CrNi 13-4	GX4CrNiMo 16-5-2	GX2CrNi 19-11	GX2CrNiMo 19-11-2	GX2CrNiMoNb 17-13-4	GX2CrNiMoN 22-5-3	GX2CrNiMoCuN 25-6-3-3	GX2NiCrMoCuN 29-25-5	G-CuSn 10	NiCu30Al	G-CuAl 10Ni
		Material-number	0.3025	0.7040	0.7660	1.0619	1.4107	1.4317	1.4411	1.4309	1.4409	1.4446	1.4470	1.4517	1.4587	2.1050.301	2.4374	2.0975.01
		UNS number	F 12401			J 03002	J 91150	J 91540		J 92500	J 92800		J 92205	J92205				
		ASTM	A 278 Class 30	A 536-60-40-18	A 439 Type D2	A216 Gr WCB	A 217 Gr CA-15	A 743 Gr CA6NM	A 743 Gr CB 7Cu2	A 743 Gr CF-3	A 743 Gr CF-3M		A 890 Gr CD3MN	A 890 Gr 4A				
Pitting index		PI	-	-	-	-	-	14	21	20	25	34	34	44	48	-	-	-
Impeller, diffuser, volute	W1 forming protective films	Qualific.	u	u	u	g	e	e	e	e	e	e	e	e	e	g	e	e
		w_{al} (m/s)	17	20	20	30	45	70	70	50	50	50	60	60	50	22	50	45
	W2 untreated water, sewage	Qualific.	u	u	u	u	e	e	e	e	e	e	e	e	e	g	e	e
		w_{al} (m/s)	15	17	17	25	45	70	70	50	50	50	60	60	50	22	50	45
	W5 & W6 Demineralized water	Qualific.	0	0	0	0	e	e	e	e	e	e	e	e	e	g	g	g
		w_{al} (m/s)	0	0	0	0	45	70	70	50	50	50	60	60	50	22	50	45
Wear rings	W1 forming protective films	Qualific.	u	u	u	u										g		
		w_{al} (m/s)	12	15	15	17										15		
	W2 untreated water, sewage	Qualific.	u	u	u	u										g		
		w_{al} (m/s)	10	12	12	15										15		
	W5 & W6 demineralized water	Qualific.	0	0	0	0										g		
		w_{al} (m/s)	0	0	0	0										15		

Qualification: 0 = not appropriate, not recommended; u = usable ; g = good; e = excellent

Note: w_{al} is the allowable velocity relative to the component (not the tip speed u_2, which is roughly twice as large for $n_q < 50$)

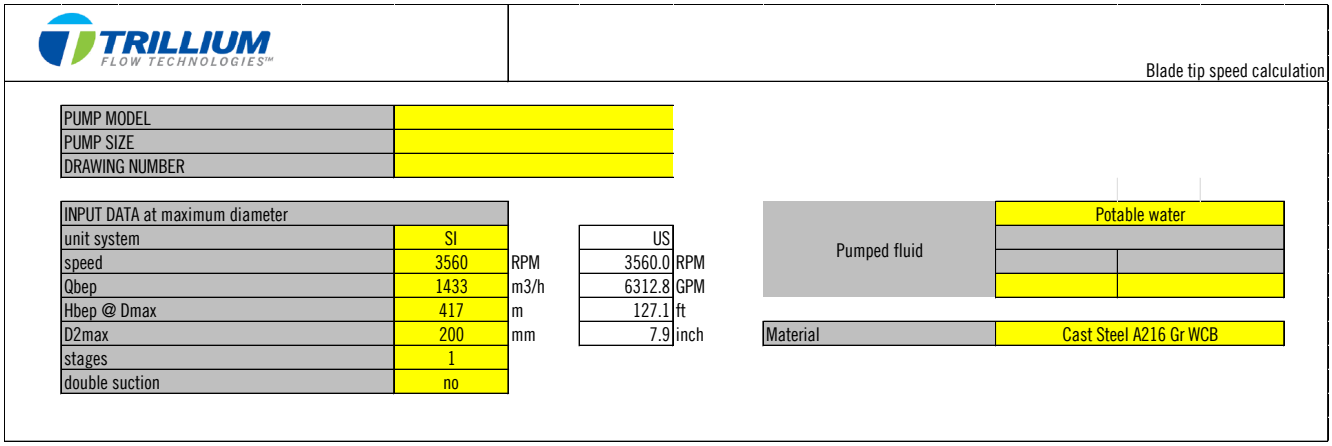
Figure 5: velocity limit for cast materials.

5. Description of the module

The screen for the input data is shown in *Figure 6*.

Only data on yellow cells must be filled in.

Data can be filled either using the SI or US unit system.



Blade tip speed calculation

PUMP MODEL	
PUMP SIZE	
DRAWING NUMBER	

INPUT DATA at maximum diameter				
unit system	SI		US	
speed	3560	RPM	3560.0	RPM
Qbep	1433	m3/h	6312.8	GPM
Hbep @ Dmax	417	m	127.1	ft
D2max	200	mm	7.9	inch
stages	1			
double suction	no			

Pumped fluid	Potable water	
Material	Cast Steel A216 Gr WCB	

Figure 6: Input data for Blade tip speed calculation.

The list of pumped fluids and impeller materials are shown in *Figure 7*:

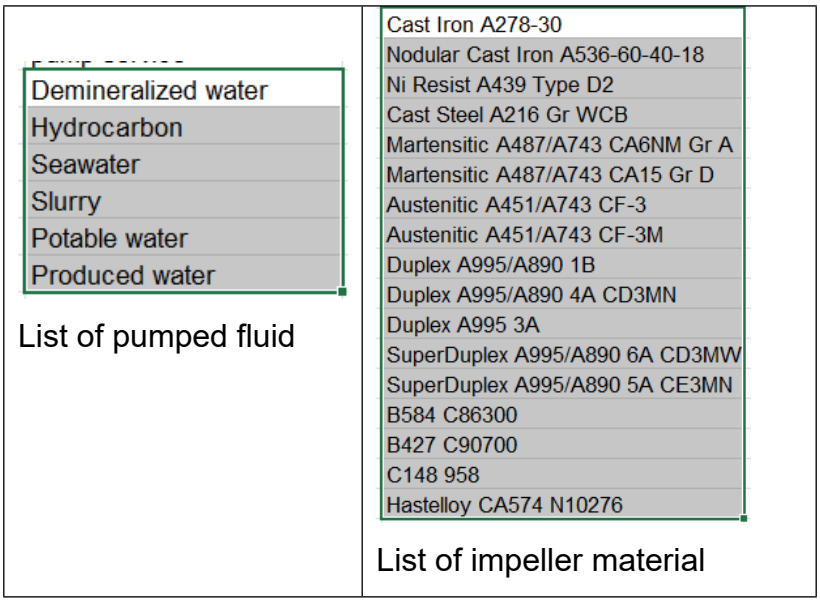


Figure 7: Available list of fluids and materials.

Output data are shown in *Figure 8*, containing:

- Specific speed of the pump.
- Maximum allowable peripheral speed as per High Energy criteria.
- Maximum allowable peripheral speed as per material and fluid selection.
- Message with results of the impeller speed limit for the input impeller diameter.

OUTPUT DATA

	SI	US
specific speed	24.3	1256.9
u2 @ specified D2	37.3	122.3

U2max based on max Energy application	122.8	402.9
U2max based on material selection and pumped fluid	0.0	0.0

Tip Speed velocity exceeds material crieria, change material or reduce the motor speed or the impeller diameter

Figure 8: Output data.

Three warning messages will be shown during the verifications:

- “Tip Speed velocity exceeds high energy criteria, reduce the motor speed or impeller diameter” if the impeller U2 > U2.al for energy criteria.
- "Tip Speed velocity exceeds material criteria, change material or reduce the motor speed or the impeller diameter" if U2 > U2.al for material selection.
- "Tip Speed velocity inside the limit" if the previous conditions are not met.

6. Acceptance criteria

The acceptance criterion of peripheral speed verification is based on 2 checks:

- U2 < U2.al for energy criteria.

AND

- U2 < U2.al for material selection criteria.

Further restrictions are used for Hydrocarbon fluids AND overhung pump type where U2.al =62 m/s.

If one of those assumptions is not verified, the impeller tip speed for the selected diameter is not verified.

7. Limitations

The method is applicable for defining and verifying the impeller tip speed due to stresses and erosion aspects.

No effects due to cavitation erosion have been considered.

In case of special fluid or special material not considered in the module, the maximum speed limit must be confirmed by Product Engineering.

8. References

[1] J.F.Gulich "Centrifugal Pumps" Fourth Edition - 2020

[2] API 610 standard